# Multi-Source Candidate Transformer

**Design Document**

**1. Project Overview**

The Multi-Source Candidate Transformer is a modular Python-based pipeline that consolidates candidate information from multiple input sources into a standardized JSON profile. The system parses data from an ATS JSON file, a resume PDF, and recruiter notes, then normalizes, merges, validates, and projects the information into a configurable output format.

**2. Objective**

The objective of this project is to:

Parse candidate data from heterogeneous sources.

Normalize extracted information into a common schema.

Merge records using configurable source priorities.

Validate mandatory fields.

Calculate confidence for extracted information.

Generate a standardized candidate JSON.

**3. System Architecture**

```
ATS JSON
   |
   ▼
 ATS Parser

Resume PDF
   |
   ▼
Resume Parser

Recruiter Notes
   |
   ▼
Notes Parser
    |
    ▼
 Normalizer
    |
    ▼
 Merger
    |
    ▼
 Confidence Calculator
```

```
     |
     ▼
 Validator
     |
     ▼
 Projection
     |
     ▼
Final Candidate JSON
```

**4. Pipeline Description**

**ATS Parser**

Reads structured candidate information from the Applicant Tracking System (ATS) JSON file.

**Output**

Candidate ID

Name

Email

Phone

Skills

Experience

**Resume Parser**

Extracts candidate details from a PDF resume using pdfplumber and regular expressions.

Extracted information includes:

Name

Contact Information

Skills

Education

Experience

Projects

Certifications

Achievements

LinkedIn

GitHub

**Recruiter Notes Parser**

Processes recruiter feedback from a text file.

Typical information includes:

Technical observations

Communication feedback

Overall recommendation

**Normalizer**

Converts extracted information into a consistent format by:

Standardizing phone numbers

Removing duplicate entries

Normalizing text fields

Formatting data according to the candidate schema

**Merger**

Combines information from multiple sources using configurable source priorities.

Example priority:

ATS

↓

Resume

↓

Recruiter Notes

**Confidence Calculator**

Computes an overall confidence score based on field completeness and successful extraction.

**Validator**

Ensures mandatory fields are available before generating the final output.

Validation checks include:

Candidate ID

Name

Email

Phone Number

Skills

**Projection**

Transforms the merged candidate object into the required JSON schema defined by the project configuration.

## 5. Project Structure

```
Multi-Source-Candidate-Transformer
|
├── config/
├── docs/
├── input/
├── output/
├── parsers/
├── pipeline/
├── tests/
├── utils/
├── validators/
├── main.py
├── README.md
└── requirements.txt
```

## 6. Technologies Used

Python 3

pdfplumber

Regular Expressions

JSON

Object-Oriented Programming (OOP)

## 7. Design Decisions

The project follows a modular pipeline architecture where each component has a single responsibility.

Parsers extract data from different sources independently.

The Normalizer ensures a common schema.

The Merger combines data based on configurable source priorities.

Validation prevents incomplete outputs.

Projection produces the final standardized JSON.

This design improves maintainability, scalability, and allows new data sources to be added with minimal changes.

## 8. Future Enhancements

Support OCR for scanned resumes.

Add DOCX resume parsing.

Introduce AI-assisted information extraction.

Expose the pipeline as a REST API.

Integrate with enterprise ATS platforms.

## 9. Conclusion

The project demonstrates a modular and extensible candidate transformation pipeline capable of consolidating information from multiple heterogeneous sources into a unified, validated, and configurable candidate profile.